Universiteit Antwerpen

Exam Computer Networks

Example Exam

Name:	Student number: ....

General remarks

- Fill out your name and student number at the top of each page.
- Place your student card visibly on your desk.
- You may bring 1 double-sided A4 sheet with notes.
- No other course material, notes, or book may be consulted during the exam.
- The use of a cell phone or any other means of communication is strictly forbidden.
- Always motivate your answer. Show how you came to your results.
- The expected size of the answer is in line with the size of the provided answer box.
- If your answer does not fit within the box, ask for an extra piece of paper. Clearly mark that your answer continues elsewhere.
- Plan your time well, you have **3 hours**.
- You can use a calculator.
- You are free to provide answers either in English or in Dutch.

Name: ………………………………………… Student number: …………………

Question 1: Store-and-forward

Most packet switches use store-and-forward transmission.

1. Explain, using a figure, how store-and-forward transmission works.

2. For a fixed packet size of L bits, and a link transmission rate of R bps, calculate the time to send a single packet over a network path from a source to a destination node with 2 intermediary routers, assuming store-and-forward transmission, but ignoring other delays such as propagation and queueing delays.

3. Now perform the same calculation but for sending N packets instead of 1.

4. Finally, generalize the calculation further, assuming a packet size of L bits, a link transmission rate of R bps, N packets being transmitted, and a path containing M intermediary routers.

Name: ………………………………………… Student number: …………………

Question 2: HTTP/2 performance

1. Briefly explain the concept of framing in the context of HTTP/2. What problem does it solve?

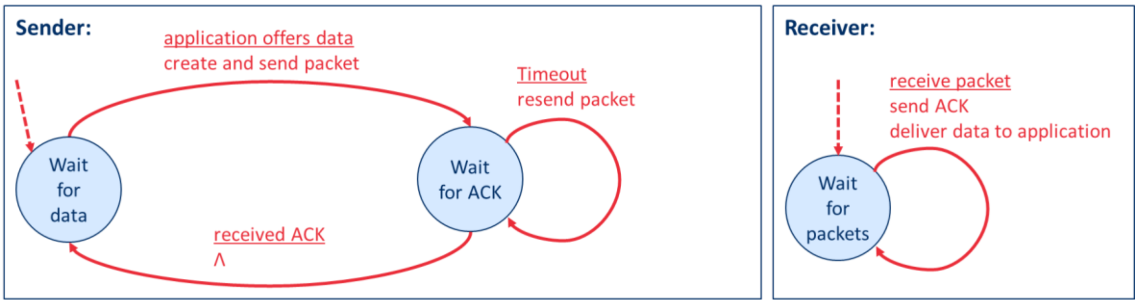
2. Consider sending a web page consisting of a video clip and five images. Suppose that the video clip is transported as 1000 frames, and each image consists of 20 frames. The HTTP client opens a single TCP connection to the server. How long (expressed in number of frame times) would it take to download the five images in HTTP/1.1 and HTTP/2 respectively, assuming the client sends a request for the video object first?

3. Now assume that the server makes use of HTTP/2 prioritization. The five images are given a priority of 50, while the video is given a priority of 20. How long would it take to download all five images now?

Name:	Student number: ....

Question 3: ARQ protocol

Given an ARQ protocol as implemented in the below finite state machine.



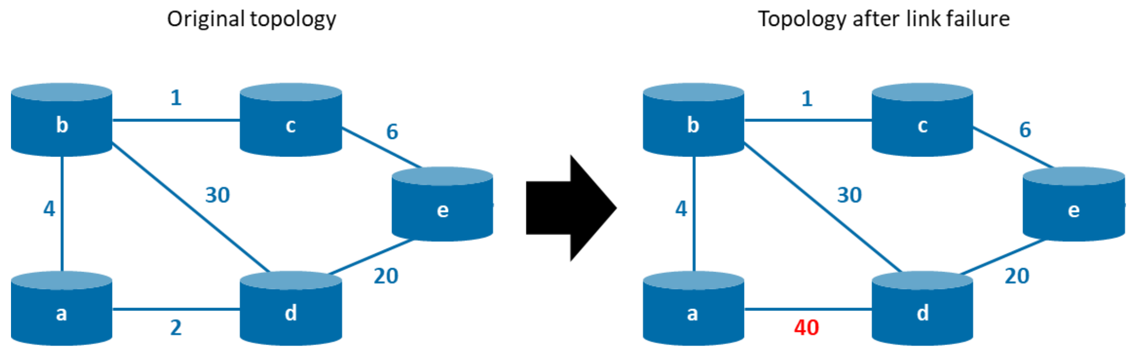
1. What are the problems that occur in the above protocol? How can they be solved?

2. Draw a new finite state machine that solves the identified issues.

Name: .... Student number: ....

Question 4: Distance vector (DV) routing algorithms

Consider the network topology below, consisting of 5 nodes (a, b, c, d, and e). The numbers represent the link costs. The figure on the left is considered the original situation. The one on the right represents a change after a link failure occurs, where the cost on the link from node a to node d increases from 2 to 40.



1. First consider the original topology (leftmost figure). What would be the information known to **node a** when initializing distance vector (DV) routing algorithm?

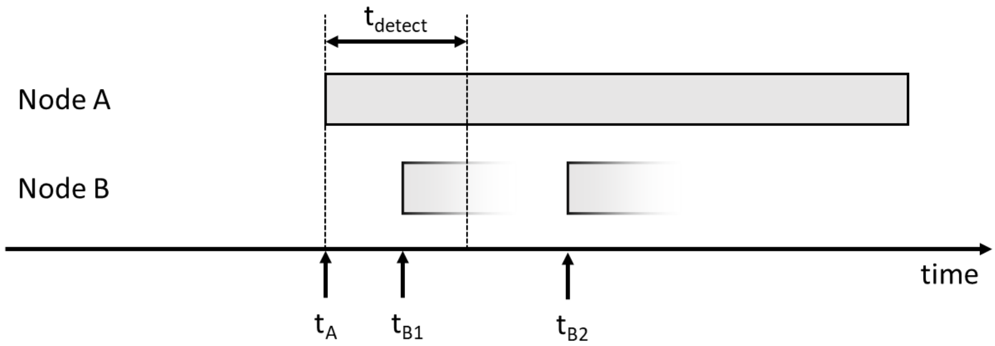
2. Considering again the original topology, what would be the information known to **node a** after the DV routing algorithm converges?

3. Now consider the situation to change to that in the rightmost figure after the DV algorithm has converged. What steps would be involved, from the point of view of **node a**, for the DV algorithm to converge once again?

4. Did you notice a problem in the behaviour of the DV algorithm when the link cost of node a to d changes? How can this problem be solved?

Question 5: Random Access Protocols

Consider the situation as shown in the figure below, where nodes A and B share a broadcast medium. Node A starts a transmission at time t_A. The time interval it takes node B to detect node A's transmission equals t_{detect}.



1. Now consider a scenario where a link layer frame becomes available for transmission at node B at time t_{B1}, with $t_A < t_{B1} < t_A + t_{detect}$. Explain what happen in case node A and B use (1) pure ALOHA, (2) CSMA/CD, and (3) CSMA/CA as multiple access protocol.

Name: ………………………………………… Student number: …………………

2. Now consider a slightly different scenario, where the link layer frame at node B becomes available for transmission at time t_{B2}, with $t_A < t_{B2} < t_A + t_{detect}$. Would the behaviour of pure ALOHA, CSMA/CD, and/or CSMA/CA be different? Explain why (not).

3. ALOHA, CSMA/CD and CSMA/CA all belong to the family of random access protocols. What is the shortcoming of these protocols? How can it be solved?